

PRINCIPLES OF MACROECONOMICS

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LECTURE 1: INTRODUCTION TO 14.02 PRINCIPLES OF MACROECONOMICS

MACROECONOMICS studies an economy as a whole, as opposed to an individual market, company, or household. Its questions concern inflation, unemployment, and exchange rates, rather than one product's price or one person's job.

1.1 Why Macro Is Not Just “Lots of Micro”

An economy contains households, firms, and markets. To understand the whole, we need to study how their decisions affect one another. One household's spending becomes another person's income; prices and expectations connect these decisions across markets.

A useful model keeps the interactions needed to answer a question and leaves out other details. Simplifying helps us see the mechanism, but the assumptions must still fit the question.

Typical questions include why unemployment rises in a recession, how interest rates affect spending, and why a worldwide shock affects countries differently.

1.2 A Macroeconomic Snapshot: Inflation after COVID-19

The U.S. economy in early 2023 is a useful motivating case. The unemployment rate rises during recessions ([figure 1](#)); the Great Recession was followed by a long labor-market recovery, while the sharp COVID-19 disruption was followed by a much faster rebound. By early 2023, unemployment was exceptionally low and wage growth was rapid, especially in accommodation and food services.

Low unemployment and faster wage growth can help workers. Macroeconomics also asks how they relate to inflation across the economy. Wage growth and inflation tend to be strongly related: since 1982, U.S. wage growth and core services inflation have moved closely together, with a correlation of about 0.81 ([figure 2](#)). The usual inflation objective for large advanced economies is roughly 2% per year, whereas U.S. readings at the time were around 6.5% to 8%. Inflation had begun to ease, but remained far above that benchmark.

Definition: INFLATION is the rate at which the overall price level rises. It is an aggregate concept: it is not the price increase of one product. Different price indexes measure this change in different ways.

The post-COVID inflation episode was broad, not uniquely American. Many economies that had previously experienced inflation near or below 2% saw it rise substantially. A strong recovery was part of the shared background, while local conditions added further effects. In Europe, for example, the war in Ukraine and the associated energy-price shock intensified inflation.

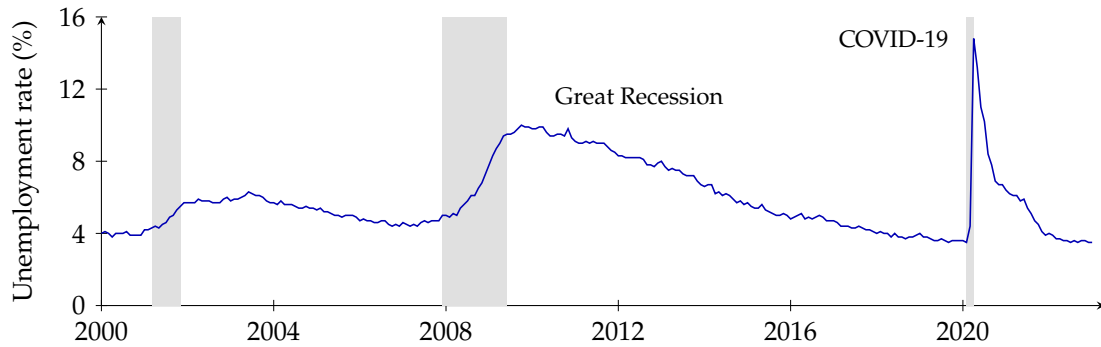


Figure 1: U.S. unemployment rate (seasonally adjusted, percent), January 2000 to January 2023 (Bureau of Labor Statistics). Shaded bands mark recessions. Unemployment took years to fall after the Great Recession but fell within about two years after its COVID-19 spike.

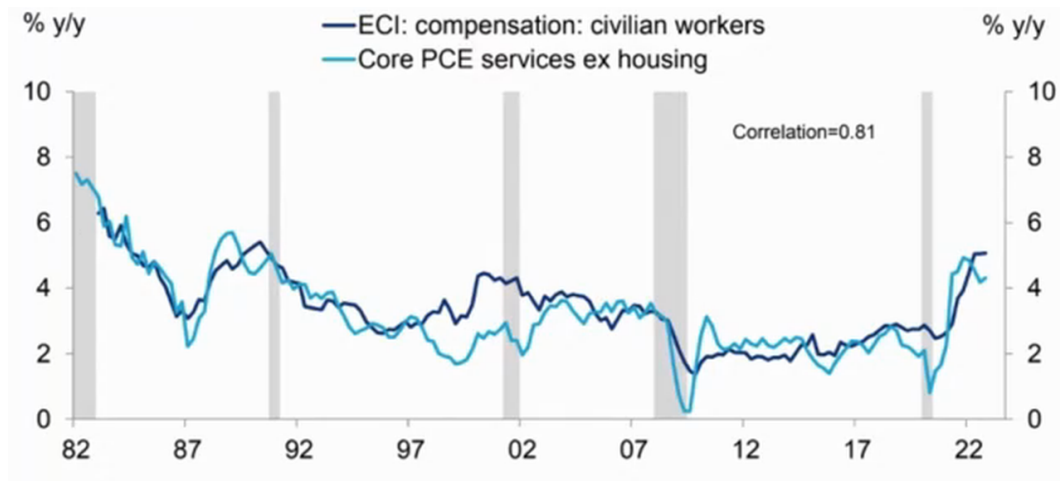


Figure 2: Year-over-year growth of the employment cost index (compensation of civilian workers) and core PCE services inflation excluding housing, 1982–2022. Shaded bands mark recessions. The two series have a correlation of 0.81 and both rose sharply after the pandemic.

Note. Numbers in this section describe the economy in early 2023; they are not current estimates. Their role is to show how macroeconomic data become a policy problem.

1.3 Interest Rates, Aggregate Demand, and Financial Markets

The central bank's main tool for controlling inflation is the interest rate. Lower interest rates make borrowing less expensive, which tends to support consumption and firms' investment. Higher rates work in the opposite direction, restraining demand. This trade-off explains the rapid rate increases undertaken once policymakers judged inflation to be too high.

Monetary policy also reaches financial markets. During the pandemic, the stock market fell sharply when COVID-19 first appeared, then rose well above its pre-pandemic level while monetary policy was very loose. Later rate hikes were associated with a broad decline in equity values. This sequence alone does not establish a complete causal story; the point is that interest rates change the discounting and financing conditions relevant to many assets.

Financial prices can move immediately when new information arrives because they reflect expectations about the future. An employment release, for example, can change investors' forecasts of inflation and future monetary policy before the stock market has even opened. These links help explain why one data release can move several markets.

Source: [MIT 14.02 Principles of Macroeconomics](#), Spring 2023, Lecture 1 video by Ricardo J. Caballero; [official transcript](#). *Figure 2* is cropped from the lecture video.

LECTURE 2: BASIC MACROECONOMIC CONCEPTS

An economy produces many different things: cars, medical care, financial services, and meals. To study the whole economy, we need measures that combine these activities. GDP, unemployment, and inflation describe output, work, and prices.

2.1 Aggregate Output and GDP

The modern national accounts provide a systematic measure of aggregate economic activity. Their central output measure is GROSS DOMESTIC PRODUCT (GDP). In these notes, “goods” is shorthand for goods *and* services: services are an important part of what households consume even when they are not tangible objects.

Definition 2.1 [GROSS DOMESTIC PRODUCT]. GDP measures the value of final goods and services produced within an economy during a period. Equivalently, it is the sum of value added in the economy. It is a FLOW: GDP for 2022 measures production over that year, not wealth at one instant.

Counting only FINAL GOODS, those bought by their ultimate users, prevents double counting. Suppose a steel producer sells steel worth \$100 to a car producer, which uses it to sell cars worth \$200 to households. The steel is an INTERMEDIATE GOOD: a good used in the production of another good. Adding the two firms’ sales would report \$300, even though the steel’s value is already included in the cars. The final-goods method therefore records GDP as \$200. This answer should not depend on whether the two firms are separately owned or are merged into one vertically integrated company.

There are three ways to reach the same total:

1. **Final-expenditure method.** Sum the market value of final goods and services produced during the period.
2. **Value-added method.** For each producer, subtract purchased intermediate inputs from sales, then add the resulting value added across producers. In the steel–car example, the steel firm adds \$100 and the car firm adds \$100, again yielding \$200.
3. **Income method.** Sum the income generated by production: compensation to labor, returns to capital, and, in a fuller account, taxes and other claims. What firms produce becomes somebody’s income.

Note. Intermediate inputs are goods or services purchased from other producers to make another good or service. A firm’s own workers and its use of its own capital are factors of

production, not purchased intermediate goods. Contracting an outside firm does not evade the accounting rule: that firm's value added is counted at its own stage of production.

The equality between aggregate production and aggregate income is especially useful in macroeconomics. In a closed economy, production creates income that people can spend on domestic output. This connection helps explain how spending and production affect one another.

2.2 Nominal and Real GDP

For final goods indexed by i , NOMINAL GDP in period t is

$$Y_t^{\text{nom}} = \sum_i p_{i,t} q_{i,t},$$

where $p_{i,t}$ is the current price and $q_{i,t}$ is the quantity produced. It is measured in current dollars, so it can rise either because the economy produces more or because prices are higher. Those are economically distinct changes.

REAL GDP holds prices fixed at a chosen base period b :

$$Y_t = \sum_i p_{i,b} q_{i,t}.$$

It therefore tracks changes in quantities at common prices. Nominal and real GDP coincide in the base period because both use that period's prices. In ordinary discussion, "GDP" usually means real GDP when the object of interest is the economy's physical output and growth.

Example: Consider an economy that produces only cars. It produces 10, 12, and 13 cars in 2011–2013, while the car price is \$20,000, \$24,000, and \$26,000. Its nominal GDP is respectively \$200,000, \$288,000, and \$338,000. Taking 2012 as the base year, real GDP is \$240,000, \$288,000, and \$312,000. The gap between nominal and real growth reflects rising prices; the real series keeps the 2012 price of \$24,000 throughout (figure 3).

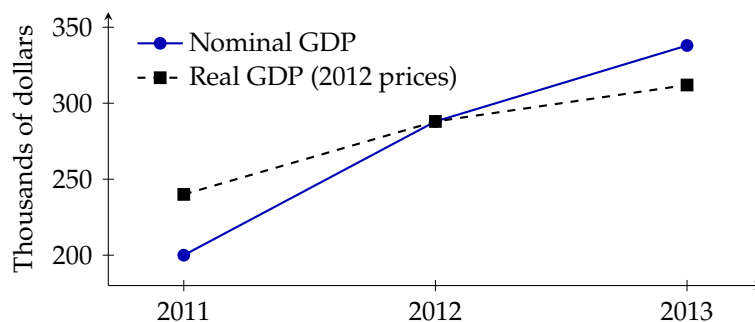


Figure 3: Nominal and real GDP in the one-good car economy, with 2012 as the base year. The series coincide in the base year; nominal GDP grows faster because the car price rises.

Using fixed base-year prices makes the idea clear. Actual GDP measurement also has to account for changing quality: a more capable computer may cost more because it is a better product, not simply because the general price level has risen. Such measurement choices matter when converting dollar values into a measure of real output.

2.3 Output Growth and Recessions

Real GDP growth describes how aggregate production changes from one period to the next. A decline in real output is a key sign of a recession, although a recession is a broader judgment about a significant and widespread downturn; the popular “two consecutive quarters of negative real GDP growth” rule is a rule of thumb, not a complete definition. The sharp 2020 contraction and subsequent reopening illustrate why growth rates should be read alongside the event that generated them.

Over long periods, much of the rise in nominal GDP can come from rising prices. From 1960 to 2018, U.S. nominal GDP increased by a factor of about 38, while real GDP increased by a factor of about 5.7. In other episodes real output may grow little or even fall while nominal GDP rises. Use real GDP when comparing production across periods with different prices.

2.4 Employment, Unemployment, and Participation

EMPLOYMENT counts people with jobs. UNEMPLOYMENT is narrower than non-employment: a person is counted as unemployed only if they do not have a job *and* are actively looking for one. Let E denote employment and U unemployment. The LABOR FORCE is

$$L = E + U,$$

and the UNEMPLOYMENT RATE is

$$u = \frac{U}{L}.$$

The denominator is the labor force, not the entire population. People without jobs who are not searching are classified as outside the labor force, so they do not enter either U or L .

In the United States, these categories are estimated with the Current Population Survey of households. Survey classifications are essential to the definition: someone who has stopped searching after a long unsuccessful spell, a DISCOURAGED WORKER, is not counted as unemployed even though their economic circumstances remain relevant. For this reason, analysts also inspect broader measures of non-employment.

The LABOR-FORCE PARTICIPATION RATE complements unemployment:

$$\text{participation rate} = \frac{L}{\text{working-age population}}.$$

It asks how many people who could in principle work are participating in the labor market. A falling participation rate can coexist with a low unemployment rate, because a smaller number of people are seeking work. In the United States, participation rose from below 60% in the 1960s to about 67% around 2000, declined after the Great Recession, and dropped sharply in 2020 (figure 4). The post-COVID recovery is a dated example: strong demand and a slower-than-expected return to the labor force both made the labor market tight. These historical examples suggest connections among jobs, output, and inflation; they do not by themselves prove cause and effect.

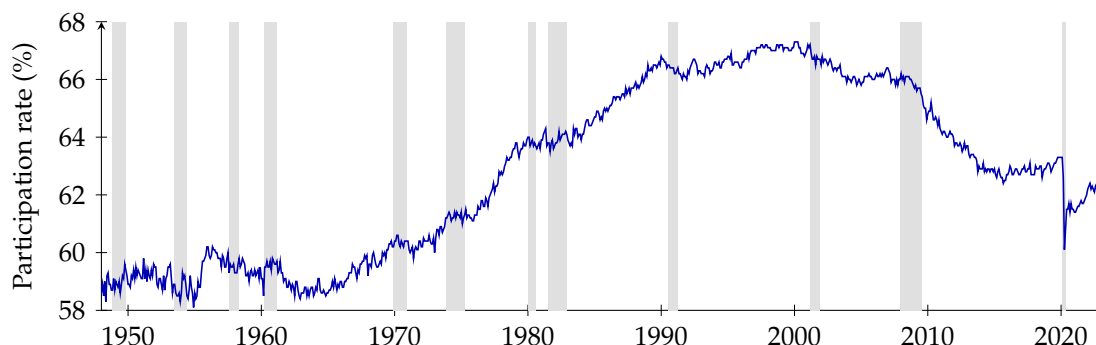


Figure 4: U.S. labor-force participation rate (seasonally adjusted, percent), January 1948 to January 2023 (Bureau of Labor Statistics), with recessions shaded. The rate rose from about 59% in the 1960s to a peak of 67.3% in early 2000, fell to about 62.5% by the mid-2010s, and dropped to 60.1% in April 2020.

2.5 Price Levels and Inflation

Definition 2.2 [INFLATION]. Inflation is a sustained rise in the general level of prices. It is not just one good becoming more expensive relative to others. If P_t is a chosen price index, its period-to-period inflation rate is

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

DEFLATION is negative inflation: a sustained fall in the general price level.

There is no single price index for every purpose. The GDP DEFLATOR is a broad economy-wide index implied by the national accounts:

$$P_t^{\text{GDP}} = \frac{Y_t^{\text{nom}}}{Y_t}.$$

It compares nominal GDP with real GDP. The CONSUMER PRICE INDEX (CPI), by contrast, follows the prices relevant to a consumer basket and is the measure most commonly reported in public discussion. The two can differ because they cover and weight goods differently, but both distinguish changes in prices from changes in real quantities.

Measures can also be tailored to a question. CORE CPI, for example, excludes food and energy because those components are often volatile. This does not make the excluded prices irrelevant; it is a way to reduce short-run noise when assessing persistent price pressure.

2.6 Why These Definitions Matter

These measures answer different questions. GDP tracks production, unemployment and participation describe the labor market, and price indexes separate price changes from changes in quantities.

Source: [MIT OCW 14.02, Spring 2023, Lecture 2](#); [official transcript](#).

LECTURE 3: THE GOODS MARKET

This model explains short-run output through spending on goods and services. It holds prices, interest rates, and productive capacity fixed so we can follow one feedback loop: spending creates production, production creates income, and income supports more spending.

3.1 The Short-Run Demand Assumption

In the very short run, firms can often vary production without immediately changing their productive capacity. The model therefore treats output as demand-determined: when planned spending on domestically produced goods changes, firms adjust production to meet it. Since production creates income, that output adjustment changes income and thereby feeds back into spending.

This is not a claim about every horizon. Over longer periods, capital, labor, technology, and prices constrain what an economy can produce. Here we focus on how a fall in demand can lower output in the short run.

3.2 Aggregate Demand

For a general open economy, AGGREGATE DEMAND is demand for domestic production:

$$Z = C + I + G + X - IM,$$

where C is consumption, I investment, G government purchases, X exports, and IM imports. Imports are subtracted because they are spending by domestic households, firms, or governments on foreign rather than domestic production.

The initial closed-economy version of the model sets $X = IM = 0$, giving

$$Z = C + I + G.$$

GOVERNMENT PURCHASES are purchases of goods and services. A TRANSFER PAYMENT, such as a check sent to a household, is not itself G : it affects demand only when the recipient changes consumption. The model takes I , G , and taxes T as given parameters; consumption supplies the behavioral link that makes aggregate demand respond to income.

3.3 Consumption and Disposable Income

DISPOSABLE INCOME is income minus taxes:

$$Y_D = Y - T,$$

where Y denotes real output and, by national-income accounting, aggregate income. The simple CONSUMPTION FUNCTION is

$$C = C_0 + C_1(Y - T), \quad 0 < C_1 < 1.$$

Definition 3.1 [AUTONOMOUS CONSUMPTION]. C_0 is consumption that does not depend on current disposable income: households would spend it whatever their level of disposable income. It represents effects we are not modeling separately, such as wealth, confidence, and expectations.

Definition 3.2 [MARGINAL PROPENSITY TO CONSUME]. C_1 is the share of an extra dollar of disposable income that is spent on consumption. If $C_1 = 0.6$, an extra dollar raises consumption by \$0.60 and saving by \$0.40.

This straight-line model is a simplification of household behavior. It makes the feedback easy to follow: higher output raises income, part of the additional income is spent, and that extra spending raises demand for output.

3.4 Goods-Market Equilibrium

Substituting the consumption function into aggregate demand yields

$$Z = C_0 + C_1(Y - T) + I + G.$$

This equation is an AGGREGATE-DEMAND FUNCTION; it describes desired spending at every possible level of output. EQUILIBRIUM adds a distinct condition:

$$Y = Z.$$

At equilibrium, production equals planned demand. Away from equilibrium, the two need not be equal: unintended inventory changes give firms a reason to alter production.

Solving the two equations gives equilibrium output:

$$Y = \frac{1}{1 - C_1}(C_0 + I + G - C_1T).$$

The term in parentheses collects the autonomous components of spending, including the tax effect on consumption. The model thus treats a change in confidence, investment, purchases, or taxes as a shift in aggregate demand, not as an immediate change in productive capacity.

Figure 5 shows the standard diagram, with income and production Y on the horizontal axis and demand Z on the vertical axis. The 45-degree line represents production, $Z = Y$. The aggregate-demand line ZZ has vertical intercept equal to autonomous spending $C_0 + I + G - C_1T$ and slope C_1 , so it is flatter than the 45-degree line when households save a positive share of an additional dollar. Their intersection A is the unique goods-market equilibrium in this simple model.

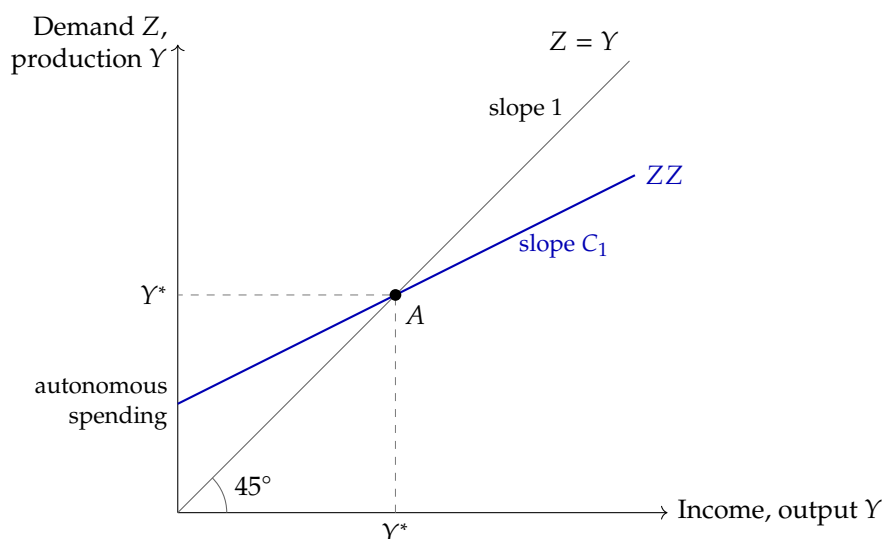


Figure 5: Goods-market equilibrium. Demand ZZ rises with income at slope $C_1 < 1$ from an intercept equal to autonomous spending. Equilibrium output Y^* is where demand equals production, at the intersection A with the 45-degree line.

3.5 The Multiplier

The coefficient

$$k = \frac{1}{1 - C_1}$$

is the **MULTIPLIER**. Because $0 < C_1 < 1$, it exceeds one. A one-unit increase in autonomous spending raises output by more than one unit:

$$\Delta Y = k \Delta A,$$

where ΔA is an autonomous shift such as a change in C_0 , I , or G .

The extra output comes from repeated rounds of spending (figure 6). For example, an initial \$1 billion increase in government purchases first raises output and income by \$1 billion. If $C_1 = 0.5$, households spend an additional \$0.5 billion; that becomes another firm's income and output, which generates a further \$0.25 billion of consumption, and so on. The geometric sequence sums to \$2 billion, precisely the multiplier $1/(1 - 0.5)$ times the initial increase.

A higher marginal propensity to consume produces a larger multiplier because more of every income increase recirculates as spending. If households save all of an additional dollar, the feedback stops after the initial round and the multiplier is one.

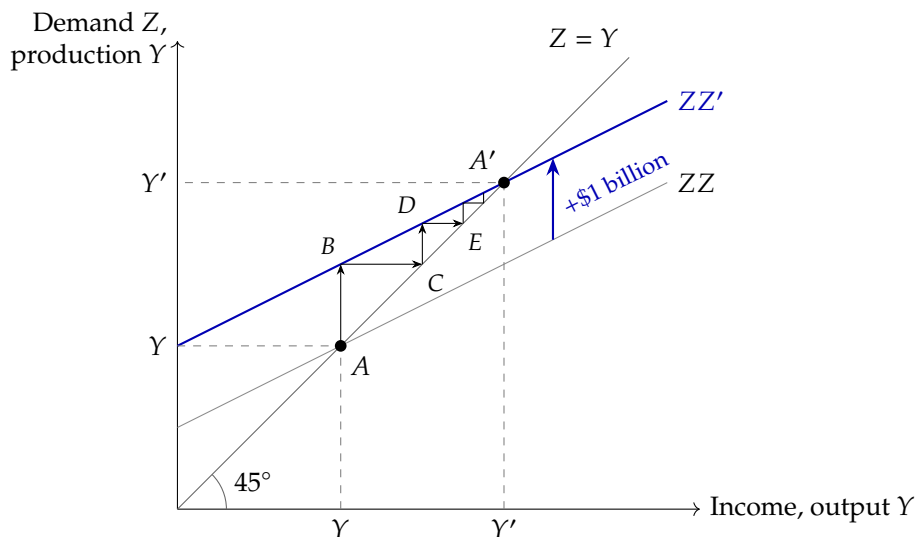


Figure 6: The multiplier with $C_1 = 0.5$. A \$1 billion rise in autonomous spending shifts demand from ZZ to ZZ' . Production rises to meet demand (A to B), the extra income raises demand (B to C to D), and the rounds shrink geometrically until output reaches A' , \$2 billion above its initial level.

Taxes affect demand indirectly through disposable income. Holding other parameters fixed,

$$\Delta Y = -\frac{C_1}{1 - C_1} \Delta T.$$

An increase in taxes reduces consumption at each level of Y and shifts the aggregate-demand line down. Its absolute effect is smaller than an equal change in government purchases in this model because households spend only the fraction C_1 of a tax change.

3.6 Saving and the IS Interpretation

The same equilibrium can be expressed using saving. PRIVATE SAVING is disposable income not consumed,

$$S_{\text{private}} = Y - T - C,$$

and GOVERNMENT SAVING is

$$S_{\text{government}} = T - G.$$

Adding the two gives total saving. Rearranging the goods-market equilibrium $Y = C + I + G$ shows that, in this closed-economy model,

$$I = S_{\text{private}} + S_{\text{government}}.$$

This is the same equilibrium written in terms of saving. It leads to the IS RELATION: the goods market is in equilibrium when investment and saving are equal.

3.7 The Paradox of Thrift

Saving can help an individual household. But if all households try to save more at once, total spending and output can fall in the short run. In this model, a stronger desire to save at every income level is a fall in C_0 . Aggregate demand shifts down (figure 7), firms reduce output, income declines, and the lower income offsets the initial attempted increase in saving until $I = S$ is restored.

The result is a PARADOX OF THRIFT: with investment and government saving fixed, a collective reduction in consumption can lower output rather than raise equilibrium saving. It is a short-run aggregate result, not advice that an individual should never save.

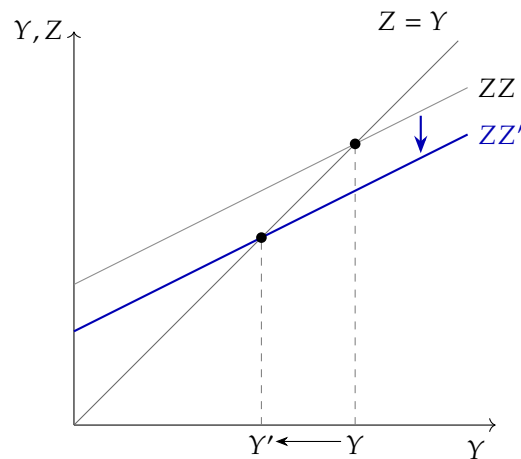


Figure 7: The paradox of thrift. A fall in C_0 shifts demand down from ZZ to ZZ' , and equilibrium output falls from Y to Y' by the multiplier times the fall in C_0 . Because I , G , and T are fixed, equilibrium private saving $S_{\text{private}} = I + G - T$ is unchanged.

3.8 Fiscal Policy

FISCAL POLICY operates through government purchases and taxes. An increase in G or a decrease in T is expansionary because it shifts aggregate demand up; the multiplier increases the total effect. A decrease in G or an increase in T is contractionary because it lowers demand and equilibrium output. Which direction is desirable depends on the economy's condition: the same demand-reducing policy that aggravates a recession can help cool an overheated economy.

Source: MIT OCW 14.02, Spring 2023, Lecture 3; [official transcript](#).